Framework vs. Library: Who's in Control?

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LIBRARY: THE TOOL

You are in control. Your code invokes library routines directly — you decide when and where to call them, so the program flow remains yours.

```
# Library call: you decide when to GET
import requests

resp = requests.get("https://api.example.com/resource")
print(resp.status_code)
```

FLOW PATTERN

Your Code → Library Utility

FRAMEWORK: THE SKELETON

The framework is in control — Inversion of Control (IoC). You register components or callbacks; the framework calls them at the right time.

```
# Framework: it calls your function
from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
  return "Hello!"
```

FLOW PATTERN

Framework Engine → Your Plugin

Compare, Choose, Combine

FEATURE	LIBRARY (TOOLBOX)	FRAMEWORK (PLATFORM)
Control	You call it.	It calls you. (IoC)
Structure	Flexible; you decide the project layout.	Strict; defines the directory and architecture.
Complexity	Focused; solved specific problems.	Broad; provides full application scaffolding.
Scope	Small, lightweight, and composable.	Heavyweight; "Batteries-included" approach.

POPULAR LIBRARIES

NumPy | Pandas | Requests | Scikit-learn | BeautifulSoup

POPULAR FRAMEWORKS

Django | Flask | FastAPI | PyTorch | Kivy

Choose the Right Tool for the Job

DECISION CHECKLIST

[v] Use a **Library** if you need specific functionality (e.g., data math, web scraping) in an existing project.

[v] Use a **Framework** if you are starting a large, standard project (e.g., a Web App) and want a proven architecture.

[!] **Hybrid Reality:** Most Python projects use both. You might build a *Django* app (Framework) and use *Pandas* (Library) inside it.

The "Golden Rule"

If you want to be the **Architect** of the program flow, pick a Library.

If you want to be a **Tenant** in a well-built, pre-configured house, pick a Framework.